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ABSTRACT

In this project we will be making an eye pupil gaze controlled wheel chair. It is used for people affected by Paralysis and Spine injuries are unable to move their body for locomotion. The product consists of an external camera that monitors the eye movement. Open CV software is used for Facial Recognition in this project. Betaface, BioID, Deep Vision AI are other software used for facial recognition.

The wheel chair consists of Raspberry pi (model 4b) Microcontroller. The obtained data is compared with the database and the wheels are moved in the respective directions with help of Microcontroller.

The Wheelchair is dependent system used by elderly and physical disable persons. Here introducing the design implementation models of totally independent Eye control electric wheelchair. As per requirement of the disabilities deferent kind of automatic systems are available in market such as voice control or joystick control system. Sometime for totally paralysis person may be have very difficult to use that type of systems. Here the Eye control system provides the independence to make their life easy and more convenient . And also they save the huge amount of energy or external man power. Camera captured the image in real time and analysis the image as input to set the commands for interface the motor driver IC through sending the commands to GPIO pins.

The motor driver circuit is used to perform the different operation such as left, right, forward and stop. For the advance level of Image Processing open computer vision (OpenCV) library is used for Face and Eye detection . And several application and algorithms are used to find out accurate pupil location detection and tracking of that. One of them is Hear cascade like features detection algorithm used to detects single or multiple face and detection of both eye . To detecting the exact Eye pupil and locate its centre point is ultimate goal of this system. For automatically find out Eye pupil and tracking eye pupil many computer vision library of Image processing are used like object detection, motion detection, Image colour conversion, edge detection, pattern matching etc.

LIST OF ABBREVIATIONS

SL. NO	ABBREVIATION	EXPANSION
1	SSH	SECURE SHELL
2	SCP	SECURE COPY PROTOCOL
3	VNC	VIRTUAL NETWORK COMPUTING
4	OPEN CV	OPEN SOURCE COMPUTER VISION LIBRARY
5	GPIO	GENERAL PURPOSE INPUT OUTPUT
6	SPI	SERIAL PERIPHERAL INTERFACE
7	I2C	INTER INTEGRATED CIRCUIT
8	UART	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER
9	PMC	PULSE CODE MODULATION
10	VS CODE	VISUAL STUDIO CODE
11	NUM PY	NUMERICAL PYTHON
12	IDE	INTEGRATED DEVELOPMENT ENVIRONMENT
13	AR	AUGUMENTED REALITY

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The Wheelchair is dependent system used by elderly and physical disable persons. Here introducing the design implementation models of totally independent Eye control electric wheelchair. As per requirement of the disabilities deferent kind of automatic systems are available in market such as voice control or joystick control system. Sometime for totally paralysis person may be have very difficult to use that type of systems. Here the Eye control system provides the independence to make their life easy and more convenient . And also they save the huge amount of energy or external man power. Camera captured the image in real time and analysis the image as input to set the commands for interface the motor driver IC through sending the commands to GPIO pins. The motor driver circuit is used to perform the different operation such as left, right, forward and stop. For the advance level of Image Processing open computer vision (OpenCV) library is used for Face and Eye detection . And several application and algorithms are used to find out accurate pupil location detection and tracking of that. One of them is Haar cascade like features detection algorithm used to detects single or multiple face and detection of both eye . To detecting the exact Eye pupil and locate its center point is ultimate goal of this system. For automatically find out Eye pupil and tracking eye pupil many computer vision library of Image processing are used like object detection, motion detection, Image colour conversion, edge detection, pattern matching etc.

1.2 EXISTING SYSTEM

There are several open-source projects and DIY guides available for building eye-controlled wheelchair using off-the-shelf components and open-source software. These projects often leverage platforms like Arduino or Raspberry pi for hardware control and may use open-source eye-tracking software. This system vary in terms of features, complexity, and accessibility. Some are commercially available products, while others may require more technical expertise to assemble and customize.

1.3OBJECTIVE OF THE PROJECT

The objective of the project is implemented for the eye controlled based independent and cost effective system. The purpose of Eye movement based control electric wheelchair is to eliminate the necessity of the assistance required for the disabled person. And it provides great opportunity of the disabled to feel of independent accessible life. The implemented system will allow the disabled person to control the wheelchair without the assistance from other persons.

1.3 ADVANTAGES OF THE PROJECT

By using this system, the disable person does not need any human help. The changes of the reflected intensity can be translated into signals to determine an eye position. After the gaze direction is

determined, it is used to steer the wheelchair accordingly. Furthermore, the chair includes safety feature that implement proximity sensor to prevent crashing into objects.

1.4 ORGANIZATION OF THESIS

The thesis comprises of four chapters followed by reference and appendices. The current chapter is the introduction which gives an overview of the project “EYE GAZE CONTROLLED WHEEL CHAIR”. It provides a brief description about the advantage of the proposed system and the shortcomings in the existing system.

Chapter 2: This chapter elaborates hardware implementation. It comprises the block diagram, overall schematic and interfacing of each part of the block.

Chapter 3: This chapter deals with software implementation. The algorithm and the working process of the project are clearly shown in the images.

Chapter 4: This chapter concludes the project with result images of the entire setup and explains the future scope of the project.

1.5 CONCLUSION

In summary, the introduction chapter has set the stage for our project, providing essential context, outlining objectives, and framing the scope of our work. By elucidating the significance of the problem addressed and introducing key concepts, this chapter has laid a solid foundation for the subsequent exploration and analysis

CHAPTER 2

HARDWARE IMPLEMENTATION

2.1 INTRODUCTION

This chapter explains the working of electronic components that are interfaced with the **EYE GAZE CONTROLLED WHEEL CHAIR SYSTEM** along with its block diagram and the schematic diagram. This will explain the system design and hardware implementation i.e. the necessary components used in this project and their specifications.

2.2 BLOCK DIAGRAM

The block diagram is divided into two sections:

1. Project System, and
2. Existing System

The project system is nothing but the project, EYE PUPIL MOVEMENT CSONTROLLED WHEELCHAIR using open CV platform. Whereas, the second section, the existing system is a simulation of the existing technology. For this we use a Raspberry pi and Visual Studio Code and Open CV software

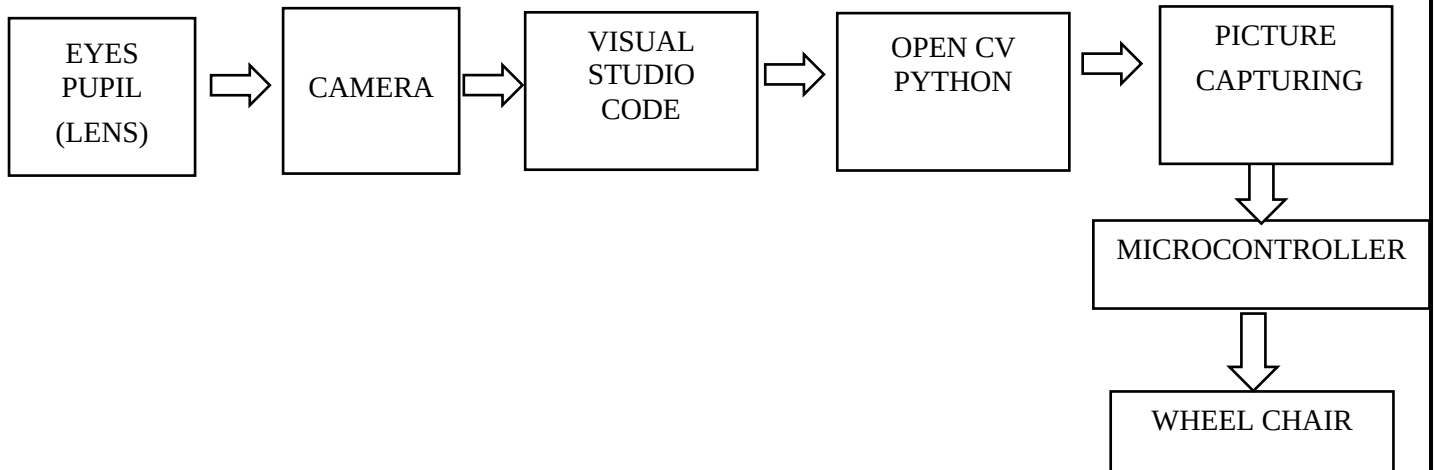


FIG 2.1 BLOCK DIAGRAM FOR PROJECT SYSTEM

2.2.1 BLOCK DIAGRAM EXPLANASION

The face is captured through the camera connected to the wheelchair. The eye pupil movement is detected by this camera. The position of the pupil is compared with the 68 facial landmark detector to find the direction of the wheels to be rotated.

The facial landmark detector finds the pupil presence in the face and compares with the already feed data in open CV written by python language. If the pupil direction is same as the database then the wheels moves to the respective directions. Raspberry PI Microcontroller controls the wheel movement by giving commands to the motor driver.

2.3 COMPONENT DESCRIPTION

The hardware components used in this project are:

- Raspberry Pi 4(Model B)
- L298 Dual H-Bridge Motor Driver
- Webcam
- 12V 100W Electric Motors
- Battery
- Jumper Wires
- Wheels
- Chain and Sprocket

The software used in this project are:

- Visual Studio Code
- Open-CV
- Python 3.8
- VNC Viewer
- Windows SCP
- Thonny
- Angry IP Scanner

The components are described in detail further.

2.4 Raspberry Pi 4(Model B)

The Raspberry Pi 4 Model B with a 1.5 GHz 64-bit quad core ARM Cortex-A72 processor, on-board 802.11ac Wi-Fi, Bluetooth 5, full gigabit Ethernet (throughput not limited), two USB 2.0 ports, two USB 3.0 ports, 1, 2, 4, or 8 GB of RAM, and dual-monitor support via a pair of micro HDMI.

2.4.1 SPECIFICATIONS OF Raspberry Pi 4(Model B)

1. **Processor:** Broadcom BCM2711, Quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.5GHz.
2. **Memory:** Options for 2GB, 4GB, or 8GB LPDDR4-3200 SDRAM (depending on the model).
3. **Connectivity:**
 - 2.4 GHz and 5.0 GHz IEEE 802.11b/g/n/ac wireless LAN.
 - Gigabit Ethernet (RJ45) with true Gigabit Ethernet throughput.
 - Bluetooth 5.0 BLE (Bluetooth Low Energy).
4. **GPIO:** 40-pin GPIO header, compatible with previous Raspberry Pi models, allowing for easy connectivity with various sensors, actuators, and other peripherals.

5. Video & Audio:

- 2 × micro HDMI ports (up to 4Kp60 supported).
- 2-lane MIPI DSI display port.
- 2-lane MIPI CSI camera port.
- 4-pole stereo audio and composite video port.

6. USB Ports:

- 2 × USB 3.0 ports.
- 2 × USB 2.0 ports.

7. Power:

- USB-C power port (supports up to 3A power input).
- PoE (Power over Ethernet) enabled (requires separate PoE HAT).

8. **Storage:** MicroSD card slot for operating system and data storage.

9. **Dimensions:** 85.6mm × 56.5mm × 17mm.

10. **Operating System Support:** Raspberry Pi OS (formerly Raspbian) is the official operating system, but it also supports various Linux distributions and even Windows 10 IoT Core.

11. Multimedia:

- H.265 (4Kp60 decode).
- H.264 (1080p60 decode, 1080p30 encode).
- OpenGL ES 3.0 graphics.

12. Other Features:

- Support for hardware-accelerated video playback.
- Support for USB booting, allowing the Raspberry Pi to boot directly from a USB mass storage device without requiring an SD card.

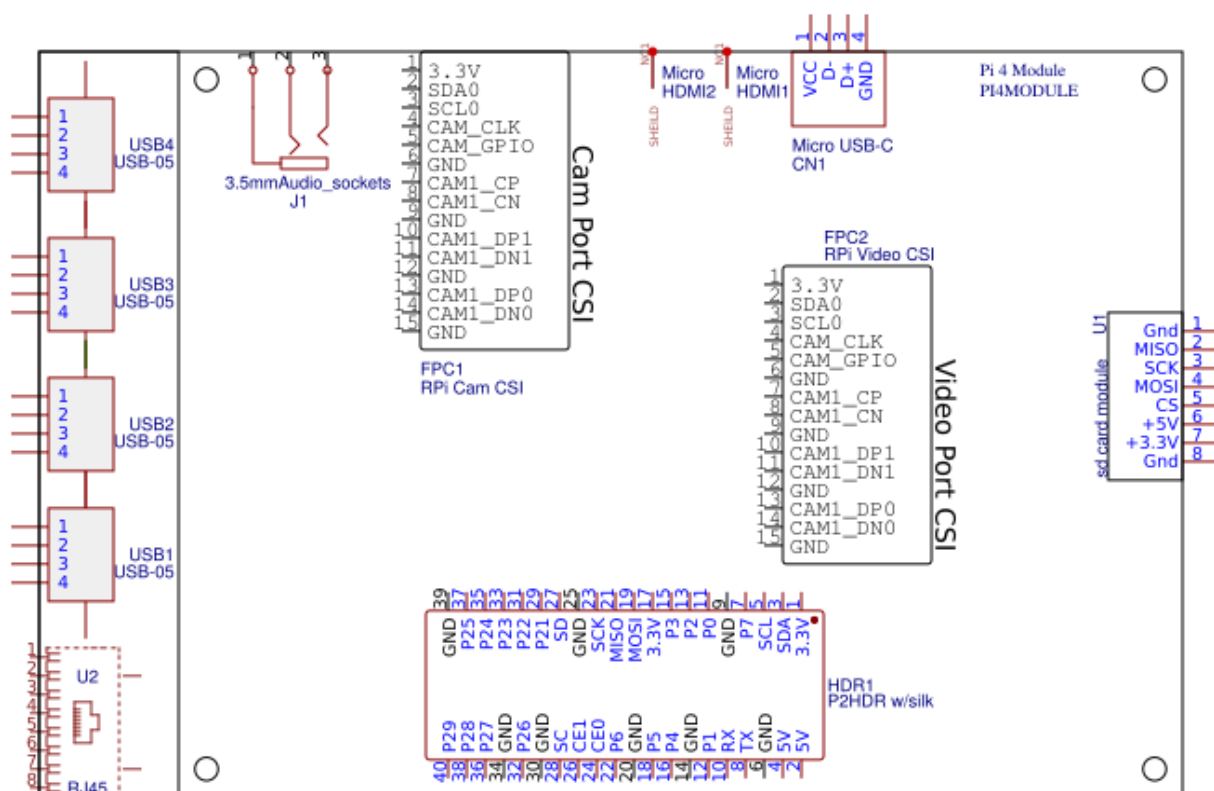


FIG 2.2 CIRCUIT DIAGRAM OF RASPBERRY PI 4(MODEL B)

2.4.2 PIN SPECIFICATION

Raspberry Pi 4 GPIO pin

Raspberry Pi 4 GPIO pins are the physical interface between the Raspberry Pi 4 and external hardware. They provide a means of communication between the Raspberry Pi 4 and external hardware. The Raspberry Pi 4 GPIO pins provide a means of communication between the Raspberry Pi 4 and external hardware.

Some pins have specific functionality, in addition to being able to be used as simple inputs or outputs. You can see them in the following table:

PWM	– Software: All pins – Hardware: GPIO12, GPIO13, GPIO18, GPIO19
SPI	– SPI0: MOSI (GPIO10); MISO(GPIO9); SCLK(GPIO11); CE0 (GPIO8), CE1 (GPIO7) – SPI1: MOSI (GPIO20); MISO (GPIO19); SCLK(GPIO21); CE0 (GPIO18); CE1 (GPIO17); CE2 (GPIO16)
I2C	– Data (GPIO2); Clock (GPIO3) – EEPROM: Data (GPIO0); Clock(GPIO1)
UART	TX(GPIO14); RX(GPIO15)

Orientate your Pi with the GPIO on the right and the HDMI port(s) on the left.

- GPIO (General Purpose IO)
- SPI (Serial Peripheral Interface)
- I²C (Inter-integrated Circuit)
- UART (Universal Asynchronous Receiver/Transmitter)
- PCM (Pulse Code Modulation)
- Ground
- 5v (Power)
- 3.3v (Power)

You can also configure a GPIO pin for a special function, such as I2C, SPI, or UART. These functions allow you to communicate with other devices using serial protocols.

GPIO (General Purpose IO) – Raspberry Pi GPIO (General Purpose IO) describes the pins on a Raspberry Pi that can be used for digital input and output. The GPIO pins can be used to control devices such as LEDs, motors, and sensors.

SPI (Serial Peripheral Interface)- SPI is a full-duplex serial protocol, meaning data can be sent and received simultaneously. SPI is a full-duplex serial protocol that can be used for communication with flash memory, sensors, real-time clocks (RTCs), analogue-to-digital converters, and more.

I2C – (Inter-Integrated Circuit) – The I2C pin is a pin that can be used to communicate with I2C devices. The I2C bus can be used to connect multiple devices, such as sensors, EEPROMs, and other I2C devices.

UART – Universal Asynchronous Receiver/Transmitter – In computing, a universal asynchronous receiver/transmitter (UART) is a computer hardware device for asynchronous serial communication in which the data format and transmission speeds are configurable.

PCM (Pulse Code Modulation) – The PCM interface is a digital interface for connecting audio devices. It is used to convert analog audio signals into digital audio signals.

5v Power – Raspberry Pi 5v power supply is a great way to get started with powering your own projects. The power supply provides 5 volts of power and is compatible with all versions of the Raspberry Pi.

3.3V Power – All Raspberry Pi models since the B+ can provide up to 500mA on the 3v3 pins

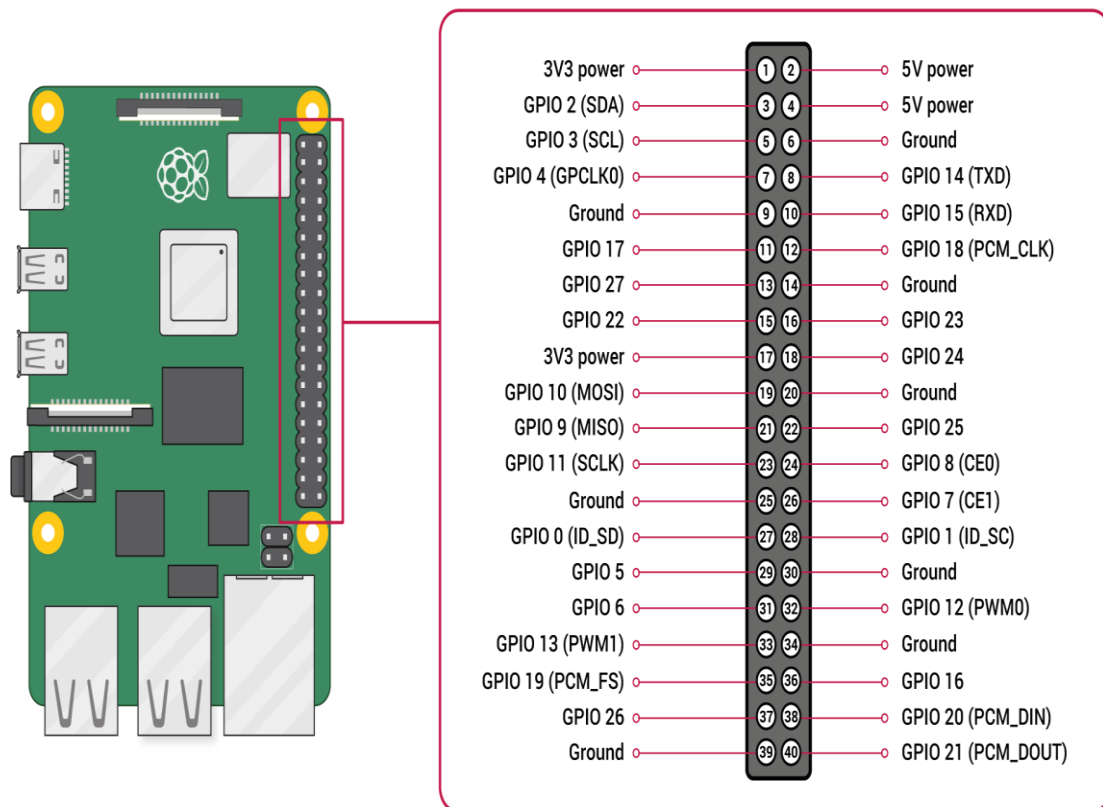


FIG 2.3 RASPBERRY PI 4 PINS

2.5 L298N Motor Driver Module

This **L298N Motor Driver Module** is a high power motor driver module for driving DC and Stepper Motors. This module consists of an L298 motor driver IC and a 78M05 5V regulator. **L298N Module** can control up to 4 DC motors, or 2 DC motors with directional and speed control.

The L298N Motor Driver module consists of an L298 Motor Driver IC, 78M05 Voltage Regulator, resistors, capacitor, Power LED, 5V jumper in an integrated circuit.

78M05 Voltage regulator will be enabled only when the jumper is placed. When the power supply is less than or equal to 12V, then the internal circuitry will be powered by the voltage regulator and the 5V pin can be used as an output pin to power the microcontroller. The jumper should not be placed when the power supply is greater than 12V and separate 5V should be given through 5V terminal to power the internal circuitry.

ENA & ENB pins are speed control pins for Motor A and Motor B while IN1& IN2 and IN3 & IN4 are direction control pins for Motor A and Motor B.

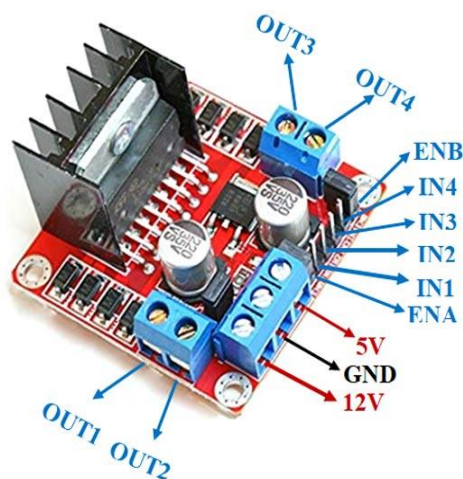


FIG 2.4.1 L298N Module PINs

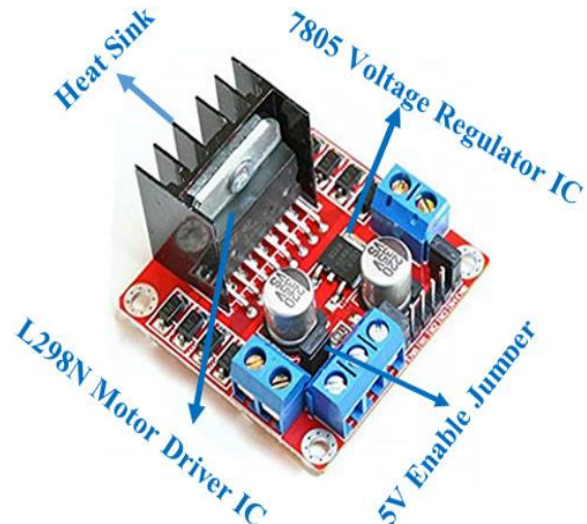


FIG 2.4.2 L298N Module

L298N Module Pin out Configuration

PIN NAME	DESCRIPTION
IN1 & IN2	Motor A input pins. Used to control the spinning direction of Motor A
IN3 & IN4	Motor B input pins. Used to control the spinning direction of Motor B
ENA	Enables PWM signal for Motor A

ENB	Enables PWM signal for Motor B
OUT1 & OUT2	Output pins of Motor A
OUT3 & OUT4	Output pins of Motor B
12V	12V input from DC power Source
5V	Supplies power for the switching logic circuitry inside L298N IC
GND	Ground pin

Features & Specifications

- Driver Model: L298N 2A
- Driver Chip: Double H Bridge L298N
- Motor Supply Voltage (Maximum): 46V
- Motor Supply Current (Maximum): 2A
- Logic Voltage: 5V
- Driver Voltage: 5-35V
- Driver Current:2A
- Logical Current:0-36mA
- Maximum Power (W): 25W
- Current Sense for each motor
- Heat sink for better performance
- Power-On LED indicator

Internal circuit diagram of L298N Motor Driver module is given below:

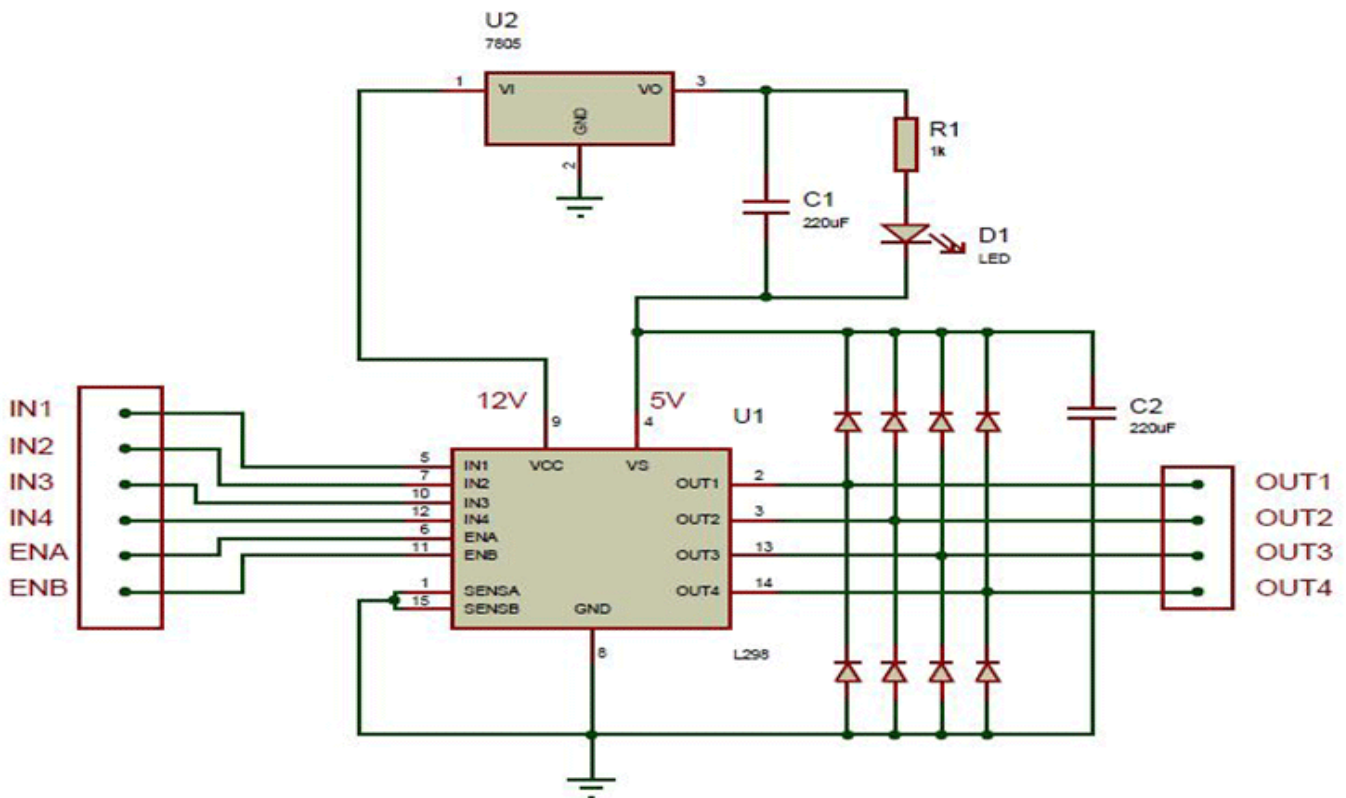


FIG 2.5 Internal circuit diagram of L298N Motor Driver module

2.6 12V 100W ELECTIC MOTOR

A **DC motor** is an electrical motor that uses direct current (DC) to produce mechanical force. The most common types rely on magnetic forces produced by currents in the coils. Nearly all types of DC motors have some internal mechanism, either electromechanical or electronic, to periodically change the direction of current in part of the motor.

A 12V 100W motor is a direct current (DC) motor designed to operate at 12 volts with a power output of 100 watts.

- **Voltage:** The motor operates on a voltage of 12 volts, which means it requires a power source that can supply this voltage to function.
- **Power Output:** The motor has a power output of 100 watts. This indicates the amount of electrical power the motor can convert into mechanical power. It determines the motor's capability to perform tasks such as driving wheels, spinning a shaft, or moving loads.
- **Type:** This motor is likely a brushed DC motor, which is one of the most common types of DC motors. It contains brushes and a commutator to deliver electrical power to the motor's windings.
- **Applications:** Such motors find applications in various fields including robotics, automotive (for small vehicles or accessories), home automation, industrial automation.
- **Size and Form Factor:** The physical size and form factor of the motor can vary based on its intended application.
- **Performance Characteristics:** Besides voltage and power, other performance characteristics such as speed, torque, and efficiency may vary depending on the specific design and construction of the motor.

- **Mounting and Connection:** Typically, these motors have standardized mounting options and electrical connections, making them relatively easy to install and integrate into various systems and projects.



FIG 2.6 12V 100W ELECTIC MOTOR

2.7 WEBCAM

A webcam is a small digital video camera that connects to a computer. It is also known as a web camera that can capture pictures or motion video. These cameras come with software that needs to be installed on the computer that helps transmit its video on the Internet in real-time. It has the ability to take pictures, including HD videos, but its video quality can be lower as compared to other camera models.



FIG 2.7 WEBCAM

Features of webcam

1. Megapixels

The megapixels are very small dots of color that makes a visual image when they are combined. Although a webcam with 320X240 or 640X480 pixels provides a better image

2. Frame Rate

As the megapixels control the image and color brightness, the frame rate deals with the video quality. At least the frame rate in a reasonable webcam will have 30 frames per second. If the frame rate is less than 30fps, this is out of date, and images may shake and vibrate.

3. Lens Quality

A moderately-priced model has a plastic lens, while a high-end webcam has a glass lens. A glass lens is much better as compared to a plastic lens.

4. Autofocus

Autofocus is a feature that works by automatically focusing the subject, while it moves around. This feature can be inappropriate that can be caused to slow functions, while the camera also takes the time to capture.

5. Resolution

A resolution is an important aspect in pictures or videos, although many webcam support 720p and 1080p high-definition quality. Also, some webcams come to have 4k capability, but they come with a premium price tag.

2.8 JUMPER WIRES

A **jump wire** (also known as **jumper**, **jumper wire**, **DuPont wire**) is an electrical wire, or group of them in a cable, with a connector or pin at each end (or sometimes without them – simply "tinned"), which is normally used To inter connect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering.



FIG 2.8 JUMPER WIRES

Types

There are different types of jumper wires

- Solid tips – The arrangement of the elements and ease of insertion on a breadboard allows increasing the mounting density of both components and jump wires without fear of short-circuits
- Crocodile clips – are used, among other applications, to temporarily bridge sensors, buttons and other elements of prototypes with components or equipment that have arbitrary connectors, wires, screw terminals, etc.
- Banana connectors – are commonly used on test equipment for DC and low-frequency AC signals.
- Registered jack (RJnn) – are commonly used in telephone (RJ11) and computer networking (RJ45).
- RCA connectors – are often used for audio, low-resolution composite video signals, or other low-frequency applications requiring a shielded cable.
- RF connectors – are used to carry radio frequency signals between circuits, test equipment, and antennas.
- RF jumper cables - Jumper cables is a smaller and more bendable corrugated cable which is used to connect antennas and other components to network cabling. Usually the most bendable jumper cable diameter is 1/2".

2.9 WHEELS



FIG 2.9 WHEELS

The wheels are constructed by a wooden hub, a fixed steel axle (the bearings were located in the fork ends), wooden spokes and a shrink fitted iron tire. A typical modern wheel has a metal hub, wire tension spokes and a metal or carbon fibre rim which holds a pneumatic rubber tire.

It consists of Hub, Axle, Bearings, Hub shell and flanges, Hub brakes, Gears, Rim.

2.10 CHAIN AND SPROCKET

A sprocket is a toothed wheel that fits onto a shaft. It is prevented from rotating on the shaft by a key that fits into keyways in the sprocket and shaft.

A chain is used to connect two sprockets. One sprocket is the driver sprocket. The other sprocket is the driven sprocket. Motion and force can be transmitted via the chain from one sprocket to another, therefore from one shaft to another. Chains that are used to transmit motion and force from one sprocket to another are called power transmission chains.

There are 6 major groups of power transmission chains:

- Standard general purpose roller chains, widely used in industry
- High performance roller chains, these roller chains are stronger than general purpose roller chains
- Lube-free chains, these chains can be used without lubrication
- Environmentally resistant chains with special corrosion resistance
- Specialty chains, Type 1. used as bicycle chains, motor cycle chains, automotive chains
- Specialty chains, Type 2 including miniature chains, leaf chains and inverted tooth chain, i.e. silent chains.

Most of these chains are the roller type, i.e. they are composed of link plates, pins that join the link plates and also rollers and bushes.



FIG 2.10.1 CHAIN AND SPROCKET

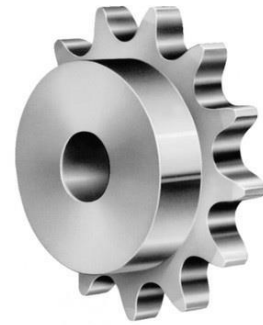


FIG 2.10.2 SPROCKET

Advantage of chain drives

An advantage of chain drives over most belt drives is that the chain cannot slip on the sprocket, so the chain and sprocket provides a positive, non-slip drive, i.e. the chain cannot slip on the sprocket because the sprocket teeth prevent the chain from slipping.

Disadvantage of chain drives

The disadvantage of the chain and sprocket drive is that it can be noisy and more expensive than a belt and pulley drive system.

CHAPTER 3

SOFTWARE IMPLEMENTATION

3.1 INTRODUCTION

In this chapter we will be discussing about the details about software implementation of the project. As we have discussed in detail about the microcontroller and various hardware used in this project. We use Visual Studio Code, Open CV, Python 3.8.

3.2 Visual Studio Code

Visual Studio Code (VS Code) is a popular source-code editor developed by Microsoft. It's known for its versatility, extensibility, and strong support for various programming languages and frameworks. With features like IntelliSense, debugging support, syntax highlighting, and Git integration, it's a favourite among developers across different platforms. Plus, its large ecosystem of extensions allows users to customize their coding experience to suit their needs. Whether you're working on a small script or a large-scale project, VS Code offers a smooth and efficient environment for coding.

Visual Studio Code is a source-code editor that can be used with a variety of programming languages, including C, C#, C++, Fortran, Go, Java, JavaScript, Node.js, Python, Rust, and Julia. It is built on the Electron framework.

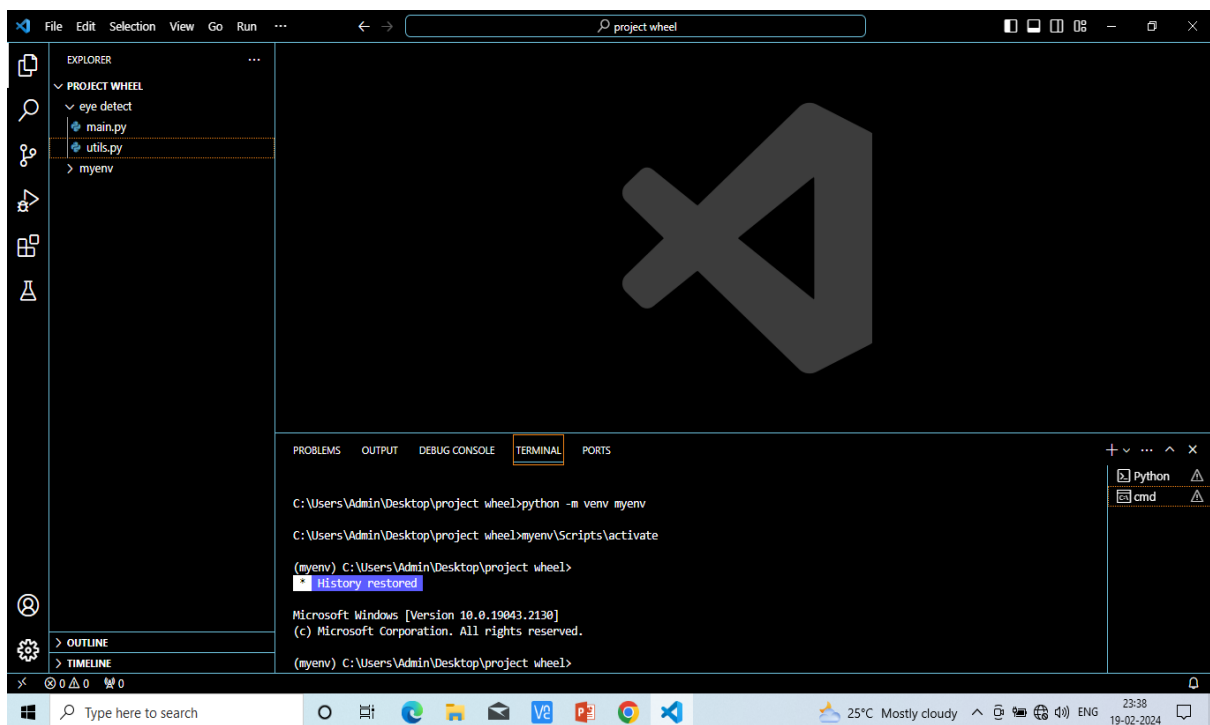


FIG 3.1 VS CODE WINDOW

3.3 OPEN CV

Open CV (Open Source Computer Vision Library) is an open source computer vision and machine learning software library. Open CV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in the commercial products.

Opencv is a huge open-source library for computer vision, machine learning, and image processing. Now, it plays a major role in real-time operation which is very important in today's systems. By using it, one can process images and videos to identify objects, faces, or even the handwriting of a human.

When it is integrated with various libraries, such as NumPy, python is capable of processing the opencv array structure for analysis. To identify an image pattern and its various features we use vector space and perform mathematical operations on these features.

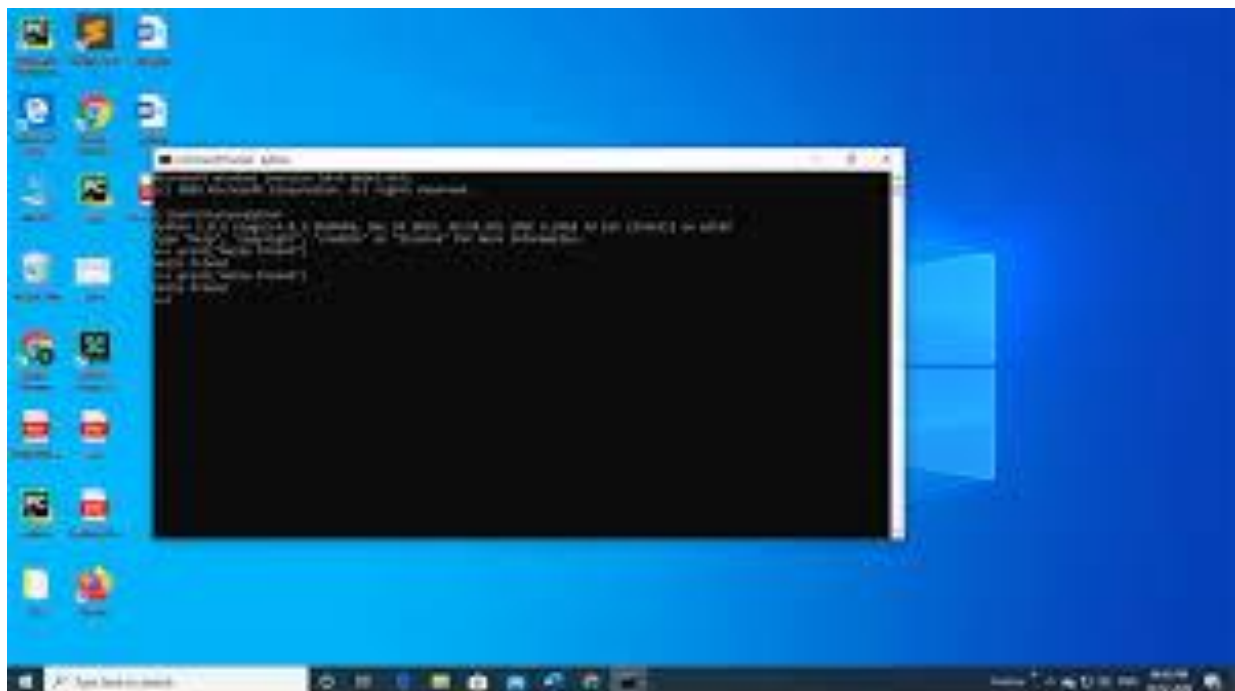
OpenCV allows you to perform various operations in the image.

- **Read the Image:** Open CV helps you to read the image for file or directly from camera to make it accessible for further processing.
- **Image Enhancement :** You will be able to enhance image by adjusting the brightness , sharpness or contract of the image. This is helpful to visualize quality of the image.
- **Object detection:** As you can see in the below image object can also be detected by using OpenCV , Bracelet , watch , patterns, faces can be detected. This can also include to recognize faces, shapes or even objects.
- **Image Filtering:** You can change image by applying various filters such as blurring or Sharpening.
- **Draw the Image:** OpenCV allows to draw text, lines and any shapes in the images.
- **Saving the Changed Images:** After processing, You can save images that are being modified for future analysis.

3.4 PYTHON 3.8

Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation.

Python is dynamically typed and garbage-collected. It supports multiple programming paradigms, including structured (particularly procedural), object-oriented and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library.



3.5 VNC VIEWER

Virtual Network Computing (VNC) is a graphical desktop-sharing system that uses the Remote Frame Buffer protocol (RFB) to remotely control another computer. It transmits the keyboard and mouse input from one computer to another, relaying the graphical-screen updates, over a network.

VNC is platform-independent – there are clients and servers for many GUI-based operating systems and for Java. Multiple clients may connect to a VNC server at the same time. Popular uses for this technology include remote technical support and accessing files on one's work computer from one's home computer, or vice versa.

VNC was originally developed at the Olivetti & Oracle Research Lab in Cambridge, United Kingdom. The original VNC source code and many modern derivatives are open source under the GNU General Public License.

VNC in KDE 3.1

There are a number of variants of VNC which offer their own particular functionality; e.g., some optimised for Microsoft Windows, or offering file transfer (not part of VNC proper), etc. Many are compatible (without their added features) with VNC proper in the sense that a viewer of one flavour can connect with a server of another; others are based on VNC code but not compatible with standard VNC.

3.6 WINDOWS SCP

SCP typically refers to the "Secure, Contain, Protect" Foundation, a fictional organization that secures and contains anomalous objects, entities, and phenomena to protect humanity. Each anomalous item, entity, or phenomenon in their containment is assigned a unique designation number and thoroughly described in SCP documentation.

Secure copy protocol (SCP) is a means of securely transferring computer files between a local host and a remote host or between two remote hosts. It is based on the Secure Shell (SSH) protocol. "SCP" commonly refers to both the Secure Copy Protocol and the program itself.

The **SCP** is a network protocol, based on the BSD RCP protocol, which supports file transfers between hosts on a network. SCP uses Secure Shell (SSH) for data transfer and uses the same mechanisms for authentication, thereby ensuring the authenticity and confidentiality of the data in transit. A client can send (upload) files to a server, optionally including their basic attributes (permissions, timestamps). Clients can also request files or directories from a server (download). SCP runs over TCP port 22 by default. Like RCP, there is no RFC that defines the specifics of the protocol.

3.7 ANGRY IP SCANNER

Angry IP Scanner is a widely-used network scanner that allows users to scan IP addresses and ports to gather information about connected devices on a network. It's a free and open-source tool available for various operating systems like Windows, macOS, and Linux.

Key features of Angry IP Scanner include:

1. **Fast Scan:** It can scan large networks quickly and efficiently.
2. **Cross-Platform:** Available for multiple operating systems, making it versatile for different environments.
3. **Customizable:** Users can specify IP ranges, ports, and other scan parameters to tailor the scanning process according to their needs.
4. **Export Functionality:** It enables users to export scan results in various formats for further analysis.
5. **Open Source:** Being open-source, it allows for community contributions and customization.

While Angry IP Scanner is primarily used for network administration and security auditing, it's important to note that network scanning should only be performed on networks and devices you have permission to scan. Unauthorized scanning of networks or devices can be illegal and unethical. Always ensure you have proper authorization before using such tools.

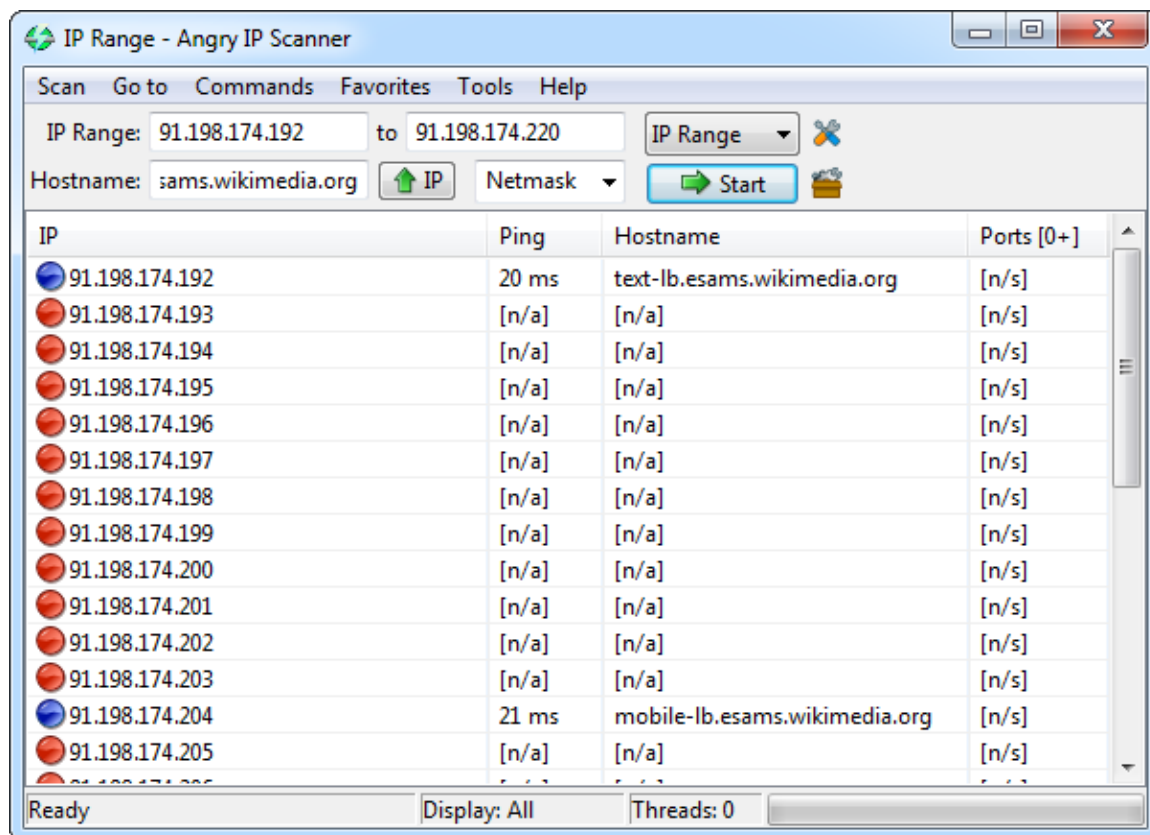


FIG 3.3 ANGRY IP SCANNER

3.8 THONNY

Thonny (is a free and open-source integrated development environment for Python that is designed for beginners. It supports different ways of stepping through code, step-by-step expression evaluation, detailed visualization of the call stack and a mode for explaining the concepts of references and heap.

Thonny is a popular Integrated Development Environment (IDE) for Python programming. It's designed with beginners in mind, offering a simple and clean interface while still providing powerful features for Python development. Some of the key features of Thonny include:

1. **Simple Interface:** Thonny has a clean and user-friendly interface that is easy to navigate, making it ideal for beginners learning Python.
2. **Code Editor with Syntax Highlighting:** Thonny's code editor provides syntax highlighting, making it easier to read and understand Python code. It also offers features like code folding and indentation guides for better code organization.
3. **Debugger:** Thonny includes a built-in debugger that allows users to step through their code line by line, inspect variables, and track down errors more easily.
4. **Variable Explorer:** It provides a variable explorer that displays the current values of variables in your code, helping you keep track of data as your program runs.
5. **Auto completion:** Thonny offers auto completion suggestions as you type, making it faster to write code and reducing the chances of typos and syntax errors.
6. **Code Execution:** Users can run their Python code directly within the IDE and view the output in the integrated shell window.
7. **Integrated Python Shell:** Thonny includes an integrated Python shell, allowing users to execute Python commands interactively and experiment with code snippets.
8. **Error Highlighting and Suggestions:** Thonny highlights syntax errors and provides suggestions for fixing them, helping users learn Python syntax and write cleaner code.
9. **Code Profiling:** Thonny includes a code profiler that can help identify performance bottlenecks in your code by measuring the execution time of different parts of your program.
10. **Virtual Environment Management:** Thonny supports the creation and management of virtual environments, allowing users to isolate project dependencies and avoid conflicts between different projects.

Overall, Thonny provides a comprehensive set of features that make it suitable for both beginners and experienced Python developers alike.

3.9 CONCLUSION

The software implementation phase has been pivotal, creating a cohesive framework that integrates seamlessly with hardware. Through iterative development, we've optimized performance and usability. Looking ahead, ongoing refinements will ensure our solution remains at the forefront of innovation, thanks to the dedication and expertise of all involved.

CHAPTER 4

RESULTS AND CONCLUSION

4.1 INTRODUCTION

From the previous chapters, all the components, software and hardware that are implemented for this project called **EYE GAZE CONTROLLED WHEEL CHAIR** is been discussed in detail. eye gaze detection wheelchair represents a ground breaking fusion of cutting-edge technology and assistive mobility solutions. At its core, this innovative device leverages sophisticated eye-tracking technology to empower individuals with severe physical disabilities to navigate their environment with newfound independence and precision. The next section will showcase more about the project's visual representation and its visual model representation.

4.2 SNAPSHOTS

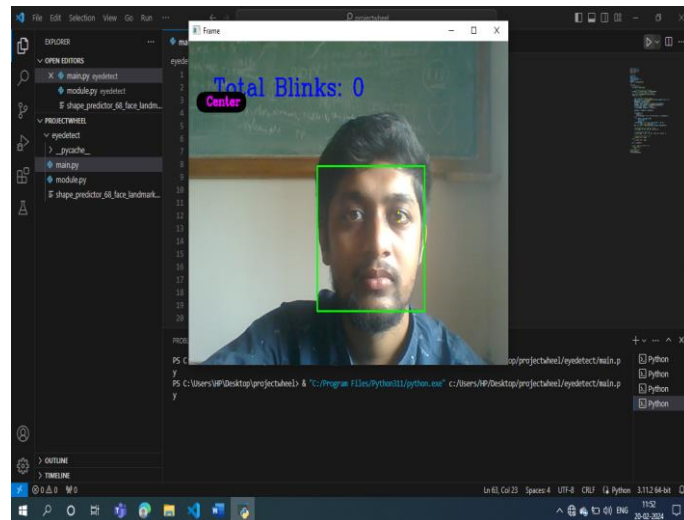


FIG 4.1 CENTER INDICATION

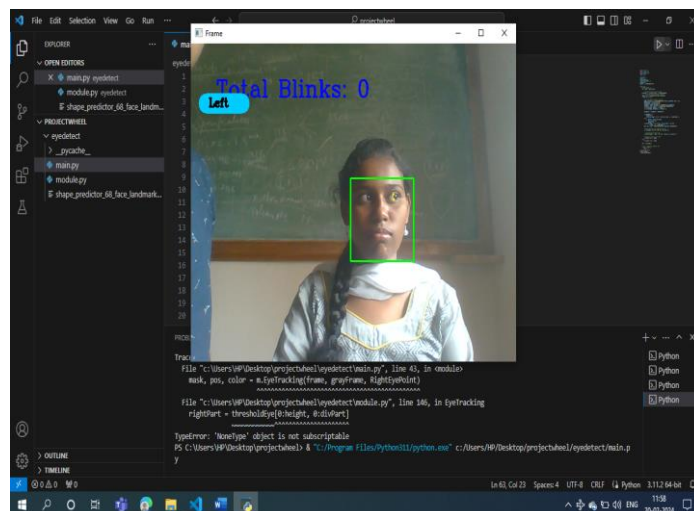


FIG 4.2 LEFT INDICATION

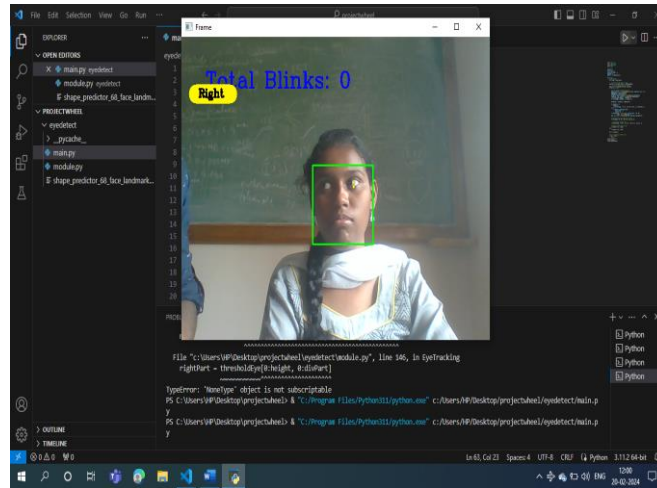


FIG 4.3 RIGHT INDICATION



FIG 4.4 FRONTAL VIEW



FIG 4.5 BACKWARD VIEW

4.3 FUTURE SCOPE

The future of eye gaze detection wheelchairs is characterized by a relentless pursuit of innovation, with the overarching goal of empowering individuals with disabilities to lead more independent, fulfilling lives. By embracing emerging technologies, fostering collaboration, and prioritizing user-centred design, the potential for transformative impact in this field is virtually limitless. Here are several potential directions in which this technology could evolve:

- **Expanded Functionality:** Eye gaze detection wheelchairs may incorporate additional features and capabilities beyond basic mobility assistance. This could include integration with smart home systems, environmental controls, or entertainment platforms, allowing users to interact with various aspects of their surroundings using only their gaze.
- **Augmented Reality (AR) Integration:** Incorporating AR technology into eye gaze detection wheelchairs could offer users enhanced situational awareness and navigation assistance.

4.4 CONCLUSION

In conclusion, the results chapter has presented the outcomes of our investigations, offering insights into the effectiveness and performance of our approach. Through rigorous experimentation and analysis, we have validated hypotheses, identified trends, and gleaned valuable findings. As we reflect on the remarkable journey of innovation that has brought us to this point, it becomes evident that the future holds even greater promise for this transformative technology. These results not only contribute to our understanding of the subject matter but also inform future research directions and practical applications.

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